Extending Distribution-Independent Estimators for the Traveling Salesman Problem to Arbitrary Dimensions

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Abstract

We present **GART 2.0**, a distribution-independent estimator of optimal TSP tour length whose feature set is dimension-agnostic (statistical in construction, applied unchanged from $d = 2$ to $d = 100$ and to non-Euclidean metrics via MDS) while feature-extraction wall time scales with d. Existing estimators either depend on convex-hull or bounding-box geometry—which is intractable for $d \geq 3$—or require the input distribution as a known model parameter. GART 2.0 does neither: it combines a deterministic Minimum Spanning Tree with a LightGBM regression on sample-statistic features (coordinate-range, centroid, and MST-topology descriptors) to predict the scaling ratio $\alpha = L_{\text{TSP}}/L_{\text{MST}}$. Because α is a bounded, unit-less ratio and the features are dimension-agnostic, the same trained model applies to $d = 2$, to $d = 100$, and—via classical multidimensional scaling on the distance matrix—to non-Euclidean metrics. The estimator's primary purpose is *precision* and *monotonicity*: small, predictable deviation from optimality—and a ranking of instances by cost that tracks the true ranking—are prerequisites for using tour-length estimates inside larger optimization pipelines, where an imprecise or rank-inverting estimate cannot be recalibrated. We evaluate GART 2.0 on this criterion rather than on absolute optimality, positioning it as a *monotone* estimator rather than a perfect one.

***K*eywords** Traveling Salesman Problem · Tour cost estimation · Distribution-independent · Multidimensional networks

1 Introduction

The Traveling Salesman Problem (TSP) is a fundamental model in combinatorial optimization, widely used for logistics operations planning, network design, and resource allocation [Chien, 1992, Applegate et al., 2006]. Because determining exact solutions for large-scale instances is NP-hard—for the general TSP by Karp [1972] and for the geometric Euclidean case specifically by Papadimitriou [1977]—the ability to compute rapid tour-cost estimates is essential for real-time decision-making in applications where a full solve is impractical. Tour-cost estimates also serve as inputs to higher-level optimization problems where the TSP appears as a subproblem. For example, in the mobile facility location problem [Halper and Raghavan, 2011], the objective function includes the cost of a TSP route through customer locations; for $n \geq 100$, replacing the exact TSP solve with a fast estimate reduces computation by one to three orders of magnitude.

1.1 Background and Motivation

Tour-length estimation is a key input to strategic logistics decisions where solving the TSP exactly is unnecessary or infeasible. Last-mile delivery accounts for approximately 53% of total shipping costs [Finmile, 2025], and the UPS ORION system is reported to save roughly 100 million miles and \$300–400 million annually through route optimization

[Holland et al., 2017]. In such systems, rapid cost estimates—rather than exact solutions—drive decisions like fleet sizing, territory partitioning, and service-level pricing. For example, a fleet planner must estimate the total route cost across hundreds of delivery zones to allocate vehicles; solving each zone's TSP exactly would take hours, while a tour-length estimator provides answers in milliseconds.

Beyond traditional logistics, fast TSP length approximation is a critical component across diverse applications in which TSP-like structures arise as subproblems. One of these application areas is bioinformatics. The ordering of markers in radiation hybrid mapping is modeled as a TSP, where rapid cost estimation enables evaluation of assembly quality without full reconstruction [Agarwala et al., 2000]. In data compression, vector quantization partitions a codebook into clusters and must order codewords to minimize transition costs, a problem reducible to TSP on a high-dimensional distance matrix. Additionally, in sensor network design, estimating the minimum data collection path length is crucial for optimizing the lifespan of battery-constrained mobile sinks in wireless sensor networks [Wang et al., 2008].

In these domains, the problem is rarely confined to 2D Euclidean space. Genome assembly operates in abstract metric spaces defined by overlap similarity [Agarwala et al., 2000], while sensor networks involve routing through high-dimensional feature spaces [Wang et al., 2008]. While a full evaluation on these application-specific metrics is beyond the scope of this study, they motivate the need for an estimator that generalizes beyond 2D Euclidean geometry. Furthermore, metaheuristics for large-scale routing frequently partition problems into smaller sub-TSPs, each requiring a rapid cost estimate to guide the search [Mariescu-Istodor and Fränti, 2021, Pan et al., 2023]. For example, Pan et al. [2023] decompose large TSP instances hierarchically, solving each cluster as an independent sub-problem. A fast, accurate tour-length estimator enables such algorithms to evaluate candidate partitions without solving each sub-TSP exactly.

Despite this breadth of applications, existing TSP tour-length estimators share a critical dependence on geometric enclosure metrics—specifically the convex hull area or bounding box volume—that fail in two important ways. First, convex hull computation scales as $O(n^{\lfloor d/2 \rfloor})$ [Chazelle, 1993], making it intractable for $d > 3$. Second, area-based formulas produce degenerate estimates on non-uniform distributions: when points lie along a diagonal line, the convex hull area approaches zero and estimators predict a near-zero tour cost, despite the true tour remaining long. Our approach addresses both limitations by replacing convex hull enclosure with a combination of MST topology and lightweight bounding-box statistics. The MST captures outlier structure and cluster separation directly from edge weights, while coordinate-range features (hypervolume, aspect ratio, centroid distances) provide scale information without requiring hull computation. We demonstrate that this approach yields lower error than all existing estimators on both synthetic and real-world TSPLIB95 benchmarks.

1.2 Limitations of the Current Models

While the need for a TSP estimation model for general cases is clear, a persistent gap remains in the literature regarding high-dimensional or non-Euclidean instances. Traditional estimators, such as the BHH theorem [Beardwood et al., 1959] and the probabilistic model by Chien [1992], were developed primarily for 2D Euclidean space. Moreover, these models commonly rely on the convex hull [Preparata and Hong, 1977, Chazelle, 1993] to estimate the "volume" or "area" of a point set. However, convex hull computation scales as $O(n^{\lfloor d/2 \rfloor})$ in arbitrary fixed dimension [Chazelle, 1993], making it prohibitive beyond three dimensions. This limitation also affects non-Euclidean metric spaces: while any n-point metric admits a low-distortion embedding into Euclidean space [$O(\log n)$ distortion, Bourgain, 1985], isometric embedding is *not* guaranteed—tree metrics and the four-point equidistant metric are standard counterexamples—and even when an approximate embedding exists, it only shifts the convex hull problem to higher dimensions rather than resolving it.

Recent work has attempted to escape these constraints with neural networks. Varol et al. [2024] report sub-1% deviation on standard distributions by combining an MLP with domain-knowledge features, but their pipeline requires solution-level features computed from heuristic tours—presupposing access to a solver. Kou et al. [2022] use the standard deviation of random feasible tour lengths—not of coordinates—as a single-feature predictor, and evaluate it on both Euclidean and non-Euclidean instances (TSPLIB95, randomly generated, and real-world inputs); the predictor is distribution-agnostic but requires sampling feasible tours, so the solver-free and feature-engineering constraints addressed here remain open. The most direct methodological neighbour is Kou et al. [2023], which pairs a Sammon-map embedding with a distribution-independent estimator on non-Euclidean TSPLIB instances; our pipeline differs by using classical metric MDS (linear, variance-retention-controlled) for the embedding while computing MST-topology features on the *original* distance matrix rather than the embedded coordinates, and by providing a single model that spans $d = 2$ through $d = 100$ Euclidean inputs on the same feature set. Neither addresses both the dimensional and the solver-free constraints simultaneously; GART 2.0 targets exactly that gap.

1.3 Contribution: The GART 2.0 Model

Throughout this work, "estimation" refers to predicting the scalar tour cost without constructing a tour, distinguishing GART 2.0 from construction heuristics and approximation algorithms that produce feasible tours with worst-case ratio guarantees. We use "distribution-free" in the sense of Cavdar and Sokol [2015]: the estimator does not require prior knowledge of the vertex distribution as an input; this does not imply uniform accuracy across all distributions (Section 5 discusses cases where error increases).

Our work contributes to the literature on distribution-independent estimators, a tradition first formalized as "distribution-free" in the 2D regression model of Cavdar and Sokol [2015]. We keep the Cavdar–Sokol design principle—estimate from sample statistics rather than from a declared distribution parameter—but replace features that scale poorly with dimensionality (convex hulls, $O(n^{\lfloor d/2 \rfloor})$; PCA [Jolliffe, 2002]) with MST-derived topological statistics and lightweight coordinate-range descriptors computable in $O(n \cdot d)$.

The simplest practical baseline for TSP estimation is the *fixed MST ratio*: $\hat{L} = \rho(d) \cdot L_{\text{MST}}$, where $\rho(d)$ is a dimension-dependent constant calibrated from empirical data (e.g., $\rho(2) = \beta_{\text{TSP}}(2)/\beta_{\text{MST}}(2) \approx 0.7124/0.663 \approx 1.075$ for uniform random instances, with $\beta_{\text{TSP}}(2)$ from Johnson et al. [1996] and the 2D MST constant from the empirical estimate of Cortina-Borja and Robinson [2000], consistent with the Avram and Bertsimas [1992] analytic lower bound); see Section 5.1.8 for the formal definition. This approach is fast, requires no geometric features, and works in any dimension. However, it applies a single global constant regardless of the instance's topological structure—clustered instances, skewed distributions, and geometric lattices all receive the same multiplier. On the 78 EUC_2D TSPLIB95 instances, this limitation produces 5.22% MAPE and 4.54% SDPE. On the diverse 2D synthetic suite described in Section 5.2, the fixed ratio's error rises to 12.45% MAPE with SDPE = 11.06% — inconsistent as well as inaccurate.

The primary contribution of this work is the Geometrically Assisted Regression Tree (GART 2.0) model, a LightGBM-based framework [Ke et al., 2017] that learns to predict a *per-instance* scaling factor $\hat{\alpha}$ rather than using a fixed constant. By combining MST-centric features—such as the leaf ratio, dominance ratio, and normalized diameter—with lightweight geometric descriptors (bounding hypervolume, centroid statistics, aspect ratio) evaluated in a canonical principal-axis frame so that the geometric block is rotation-invariant, GART 2.0 adapts its estimate to the topological structure of each instance. We frame GART 2.0 as a *precision-first*, *monotone* estimator rather than a perfect one: the goal is not pointwise optimality but a prediction whose errors are consistent across instances (low SDPE) and whose ranking of instance costs tracks the true ranking. Both properties are prerequisites for using tour-length estimates inside larger optimization pipelines, where a precise-but-biased estimator can be recalibrated by a single scaling constant and a rank-preserving estimator can drive instance-selection and partitioning decisions, but an imprecise or rank-inverting estimator cannot be corrected post hoc. On 2D synthetic data, GART 2.0 achieves SDPE = 5.19% vs. 11.06% for the MST Ratio (2.1× tighter); on multi-dimensional data, SDPE = 1.28% vs. 6.84% (5.3× tighter); and on TSPLIB95 EUC_2D, SDPE = 3.42% vs. 4.54% for the strongest classical baseline. Crucially, GART 2.0 retains the computational advantage of MST-based estimation: prediction requires only the MST (computable in $O(n \log n)$ for 2D via Delaunay triangulation, $O(n^2 \cdot d)$ in general) plus a constant-time LightGBM inference step, yielding sub-second predictions for instances where exact solvers require minutes to hours.

1.4 Paper Outline

The remainder of this paper is structured as follows. Section 2 introduces the theoretical foundations of the multiplicative MST–TSP relationship and the Christofides–Serdyukov bound [Christofides, 1976, Serdyukov, 1978]. Section 3 describes GART 2.0's feature set, dataset generation, and implementation. Section 4 presents model training with hyperparameter tuning via Optuna [Akiba et al., 2019] and complexity analysis. Section 5 defines the baseline estimators, introduces the three benchmark datasets, reports per-dataset results, and discusses the patterns they reveal. Section 6 extends GART 2.0 to non-Euclidean TSPLIB95 instances via MDS embedding. Section 7 summarizes contributions, limitations, and future directions.

2 Theoretical Foundations

To formally situate the estimation problem, we first define the Traveling Salesman Problem (TSP) using its classic integer linear programming (ILP) formulation. Let $G = (V, E)$ be a complete undirected graph where $V = \{1, \ldots, n\}$ is the set of vertices and $E = \{(i, j) : i, j \in V, i < j\}$ is the set of edges. Associated with each edge (i, j) is a non-negative cost c_{ij}, typically representing the Euclidean distance between points in $\mathbb{R}^d$. The objective is to find a tour of minimum total cost that visits every vertex exactly once. Following the standard Dantzig-Fulkerson-Johnson formulation found in Applegate et al. [2006], we use binary decision variables x_{ij}, where $x_{ij} = 1$ if edge (i, j) is included in the tour, and 0 otherwise. The problem formulation is as follows:

$$\min \sum_{i=1}^{n} \sum_{j=i+1}^{n} c_{ij} x_{ij} \tag{1}$$

Subject to the degree constraints:

$$\sum_{j<i} x_{ji} + \sum_{j>i} x_{ij} = 2, \quad \forall i \in V \tag{2}$$

And the subtour elimination constraints:

$$\sum_{i \in S} \sum_{j \in S, j>i} x_{ij} \leq |S| - 1, \quad \forall S \subset V, 2 \leq |S| \leq n-1 \tag{3}$$

The TSP is NP-hard [Karp, 1972]. This formulation establishes the combinatorial structure we exploit: any feasible tour is a 2-regular spanning subgraph, of which the MST is the cheapest connected relaxation.

In contrast to the intractable TSP, the Minimum Spanning Tree (MST) problem, which seeks a connected, acyclic subgraph minimizing the total edge weight, can be solved optimally in polynomial time. Classical algorithms such as Prim's [Prim, 1957] and Kruskal's construct the MST in $O(n^2)$ time for complete graphs. The pairwise distance computation scales as $O(n^2 \cdot d)$, making the total cost linear in d and thus a computationally stable baseline for high-dimensional inference.

The relationship between these two problems provides a theoretical bound on the optimal TSP tour cost, denoted L_{TSP}. Since any valid TSP tour is a connected graph spanning all vertices, removing one edge yields a spanning tree. Thus, the weight of the MST, L_{MST}, is a strict lower bound. Conversely, a valid tour can be constructed by traversing the MST edges twice (a depth-first walk) and applying shortcuts via the triangle inequality, establishing the classic double-tree bound:

$$L_{\text{MST}} \leq L_{\text{TSP}} \leq 2 \cdot L_{\text{MST}}. \tag{4}$$

Christofides [1976], and independently Serdyukov [1978], improved this upper bound by augmenting the MST with a Minimum Weight Perfect Matching (MWPM) on the set of vertices with odd degrees. This yields a tour whose length is guaranteed to be at most $\frac{3}{2}L^*_{\text{TSP}}$, where L^*_{TSP} denotes the optimal tour cost, in metric spaces; the algorithm is therefore increasingly cited as Christofides–Serdyukov. The ratio was recently improved to $(\frac{3}{2} - \epsilon)$ by Karlin et al. [2021], though the practical improvement is negligible. However, calculating the MWPM requires $O(n^3)$ time, which is significantly more expensive than the $O(n^2)$ MST construction. For large-scale auxiliary subproblems where computation speed is critical, the cubic complexity of the Christofides refinement is often prohibitive, necessitating a faster method to approximate the tightening coefficient.

Leveraging the relationship between the optimal TSP and MST values, instead of explicitly constructing the matching, we treat the tightening coefficient as a statistical parameter $\alpha = L_{\text{TSP}}/L_{\text{MST}}$. While the worst-case bound for α from the double-tree argument is 2.0 (Equation 4), the Christofides heuristic empirically tightens practical values. The expected value of α depends on both the dimensionality d and the instance size n. Percus and Martin [1996] showed that for uniformly distributed points, α decreases as d increases: the MST and TSP lengths converge because higher-dimensional spaces offer more "shortcuts" for the tour to exploit relative to the tree. This implies that the computational gap between the MST and the optimal TSP tour diminishes as spatial freedom increases. Consequently, a static estimator calibrated for 2D (where $\alpha \approx 1.075$ as the ratio $\beta_{\text{TSP}}(2)/\beta_{\text{MST}}(2)$ of the classical BHH constant [Johnson et al., 1996] and the empirical 2D MST constant [Cortina-Borja and Robinson, 2000]) will systematically overestimate tour costs in higher dimensions. Our proposed model addresses this by learning to predict α dynamically based on the topological characteristics of the vertex set.

3 Methodology: The Geometrically Assisted Regression Tree

We propose the Geometrically Assisted Regression Tree (GART 2.0), a framework that estimates TSP tour cost using the relationship between the MST and the optimal tour. GART 2.0 integrates gradient-boosted regression trees with a dimension-agnostic feature set derived from the MST topology to overcome the dimensional limitations of classical geometric estimators. Unlike traditional approaches that map coordinate-dependent metrics directly to tour cost, GART 2.0 decouples the estimation into two phases: (1) the deterministic computation of the MST, and (2) the statistical prediction of the scaling coefficient $\alpha = L_{\text{TSP}}/L_{\text{MST}}$. The final estimator is $\hat{L}_{\text{TSP}} = \hat{\alpha} \cdot L_{\text{MST}}$, where $\hat{\alpha}$ is predicted by a LightGBM regressor based on 30 topological features of the instance.

We constrain the regression target to $\alpha \in [1.0, 2.0]$. The lower bound is feasibility-exact: every TSP tour is a spanning structure, so $L_{\text{TSP}} \geq L_{\text{MST}}$. The upper bound is the double-MST worst-case ratio, and in all training data we observe α well inside this interval. Predictions are therefore clipped to $[1.0, 2.0]$ at inference; this acts only as a safety clamp for pathological instances where the MST approximation is degenerate (the regime outside the model's assumed operating conditions). The following section introduces the feature set.

3.1 Feature Selection

To predict the scaling factor α with low SDPE across varying dimensions, we extract 30 features from each TSP instance. The feature count reflects the need to characterize three distinct aspects of instance structure—spatial spread, edge weight distribution, and branching topology—each requiring multiple statistics to capture the range of behaviors observed across distributions. While prior estimators such as Cavdar and Sokol [2015] use coordinate-based statistics tied to specific axes (e.g., standard deviations of x and y positions) and convex hull area, our features combine MST edge weights and degree sequence—which are invariant to rotation, translation, and coordinate system choice—with lightweight geometric descriptors (bounding hypervolume, centroid distances) that generalize naturally to arbitrary d. The dimensionality d enters the feature set as an explicit input rather than implicitly through coordinate geometry.

Before any of the geometric descriptors are evaluated, we rotate the input coordinates into their principal-axis frame: the covariance matrix of the centred coordinates is diagonalised by a symmetric eigendecomposition, the eigenvectors are ordered by descending variance, and the sign of each axis is fixed by the coordinate of largest absolute magnitude so the frame is deterministic under reflections. This is the d-dimensional analogue of aligning a 2D point set with the minimum-area bounding rectangle, and it costs $O(nd^2 + d^3)$—dominated by the MST construction it precedes. The rotation leaves the MST itself invariant, because every edge weight is a Euclidean distance and Euclidean distances are preserved under rotation; it also leaves the centroid dispersion statistics invariant, because those are functions of distances to the mean. What it changes is the axis-aligned bounding-box block (hypervolume, node density, aspect ratio): in the canonical frame these descriptors no longer carry orientation noise from whatever frame the instance happened to arrive in, so two datasets that differ only by a rigid rotation now produce identical feature vectors. The entire downstream pipeline—feature extraction during training, feature extraction at inference, and every estimator that consumes the coordinates, including the classical baselines—operates on the canonicalised coordinates, so the training distribution and the test distribution share a single geometric convention.

All features can be computed in $O(n^2 \cdot d)$ time or less, dominated by the pairwise distance matrix required for MST construction. No feature requires convex hull computation or geometric partitioning; the geometric descriptors use coordinate ranges computable in $O(n \cdot d)$. We organize the features into three groups: (1) geometric and centroid descriptors, which quantify spatial spread and density; (2) MST edge distribution statistics, which capture local uniformity through edge weight moments; and (3) topological structure and clustering proxies, which detect higher-order patterns such as clusters and filaments.

3.1.1 Geometric and Centroid Descriptors

The first category quantifies the macroscopic spatial properties of the node set in $\mathbb{R}^d$. These features provide the model with scale information analogous to the volume term V in the BHH formula $\hat{L} = \beta_d \cdot n^{(d-1)/d} \cdot V^{1/d}$ [Beardwood et al., 1959], but computed from coordinate ranges rather than convex hull volume. Centroid distance statistics, adapted from Cavdar and Sokol [2015], measure the dispersion of nodes relative to the geometric center, enabling the model to distinguish uniform distributions from centralized Gaussian clusters. Table 1 lists these features.

Table 1: Geometric and Centroid Feature Specifications.

Feature Name	Description & Rationale	Complexity	Origin / Citation
Dimension (d)	The number of spatial coordinates defining each point.	$O(1)$	Standard
Number of Customers (n)	The total count of vertices in the instance.	$O(1)$	Standard
Aspect Ratio	Ratio of the largest dimension range to the smallest. Detects subspace distortion.	$O(n \cdot d)$	**Proposed**
Bounding Hypervolume	Product of coordinate ranges: $\prod(\max_d - \min_d)$. Approximates volume V for BHH bounds.	$O(n \cdot d)$	Adapted from Beardwood et al. [1959]
Node Density	Ratio of n to Bounding Hypervolume. Proxy for point intensity $\delta = n/V$.	$O(1)$	**Proposed**; cf. Beardwood et al. [1959]
Centroid Dist (Mean, Std)	Average and deviation of distances to the centroid. Measures central tendency.	$O(n \cdot d)$	Adapted from Cavdar and Sokol [2015]
Centroid Dist Max	Radius of the enclosing sphere.	$O(n \cdot d)$	Adapted from Cavdar and Sokol [2015]
Centroid Dist IQR	Interquartile range of centroid distances. Robust dispersion metric.	$O(n \cdot d)$	**Proposed**; extends Cavdar and Sokol [2015]

3.1.2 MST Edge Distribution Statistics

The second category captures the local uniformity of the point set through the statistical moments of the MST edge weights. Unlike coordinate-based variance, edge weights are invariant to rotation and translation. High skewness or kurtosis in this distribution typically indicates a mixture of scales, such as dense clusters separated by voids, which significantly impacts the α ratio.

Table 2: MST Edge Statistic Specifications

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Edge Max	Weight of the longest edge in the MST.	$O(n)$	Standard
MST Quantiles	10%, 25%, 50%, 75%, 90% percentiles. Provides a non-parametric distribution profile.	$O(n \log n)$	**Proposed**; via MST [Prim, 1957]
MST Total Length	Sum of all edge weights. The primary scaling factor; $L_{\text{TSP}} \geq L_{\text{MST}}$.	$O(n^2)$	Prim [1957]
MST Edge Mean	Average edge length in the MST.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Edge Std	Standard deviation of edge lengths in the MST.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Edge Skewness	Asymmetry of edge distribution. High positive skew implies many short edges and few long bridges.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Edge Kurtosis	Tailedness of distribution. Indicates the presence of extreme outliers.	$O(n)$	Smith-Miles and van Hemert [2010]

3.1.3 Topological Structure and Clustering Proxies

The final category contains features designed to detect higher-order structures—such as clusters, filaments, and hubs—without performing expensive clustering algorithms. We introduce specific ratios (Dominance, Gap, Normalized Diameter) that serve as $O(n)$ proxies for these complex geometric properties. For instance, the "Leaf Ratio" effectively distinguishes between low-dimensional linear manifolds (low leaf count) and high-dimensional isotropic noise (high leaf count).

Table 3: Topological and Clustering Feature Specifications

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Dominance Ratio	Sum of largest $\sqrt{n}$ edges / Total Length. Captures weight of inter-cluster bridges.	$O(n)$	**Proposed**; via MST [Prim, 1957]
MST Gap Ratio	Max edge / Median edge. Measures magnitude of sparsity jumps.	$O(1)$	**Proposed**; via MST [Prim, 1957]
MST Leaf Ratio	Fraction of nodes with degree $k = 1$. Topology proxy for dimensionality.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Degree (Mean, Std)	Branching factor statistics.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Degree Max	Maximum node degree. Identifies "hub" nodes common in High-D spaces.	$O(n)$	Smith-Miles and van Hemert [2010]
MST Diameter	Sum of edge weights on the longest path between any two leaves, computed via double-BFS.	$O(n)$	Graph Theory
Normalized Diameter	Diameter / Total Length. Scale-invariant shape proxy ($= 1.0$ for path-manifolds exactly).	$O(1)$	**Proposed**; via double-BFS
Large Edge Count	Count of edges $> \mu + \sigma$. Secondary cluster indicator.	$O(n)$	**Proposed**; via MST [Prim, 1957]

3.2 Dataset Generation

To train a general model, we constructed a synthetic dataset with topological diversity across four distribution classes: Uniform, Normal (Gaussian), Clustered (k-means centers), and Correlated (autoregressive across dimensions). For each instance, dimensions are independently assigned a distribution class, producing anisotropic configurations where the topology varies by axis. This "mix-and-match" strategy avoids bias toward any single geometric type. Random seeds are derived from distribution signatures to ensure distinct realizations across instances.

The full feature corpus comprises 106,272 solved instances with node counts $n \in [5, 1000]$ and dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ (17 training-visible dimension values), plus a held-out extrapolation bucket at $d = 100$. Instances are partitioned into training (69,768), validation (19,584), and test (16,920) via a stratified split on $(d, n,$ grid size); all $d = 100$ instances are locked into the test partition so the model never sees that dimension during training or validation. The full training grid is given in Appendix B (Table 13); extrapolation to $d = 100$ is evaluated in Section 5.3.1.

For each instance, the best-known tour cost was computed by running both Concorde [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017], retaining the shorter result as ground truth. Concorde provides provably optimal solutions for instances within its computational budget; LKH-3 is a state-of-the-art heuristic that empirically matches or approaches the optimum. For all training instances ($n \leq 1000$), Concorde confirmed optimality within its computational budget; LKH-3 matched or improved all Concorde solutions. Coordinate scaling was applied to preserve floating-point precision in the integer-valued TSPLIB format required by these solvers.

3.3 Implementation

The GART 2.0 pipeline is implemented in Python using SciPy [Virtanen et al., 2020] for Delaunay triangulation [Delaunay, 1934] and MST construction (via `scipy.sparse.csgraph.minimum_spanning_tree`), Numba [Lam et al., 2015] for JIT-compiled centroid distance computation, and scikit-learn [Pedregosa et al., 2011] for classical MDS embedding of non-Euclidean distance matrices. For instances without Euclidean coordinates (e.g., TSPLIB GEO and EXPLICIT types), MDS projects the distance matrix into 2D coordinates for geometric feature extraction while MST features are computed from the original distances; details are given in Appendix C and evaluated in Section 6. The LightGBM model [Ke et al., 2017] is trained and served via the Python API. All benchmarks were run on a single machine (AMD Ryzen 9 5900X, 64 GB RAM, Windows 11) with no GPU. Concorde version 03.12.19 and LKH-3 version 3.0.13 were used for ground-truth computation.

4 Estimation Model Development

The relationship between topological features and the scaling factor α is non-smooth: sharp regime transitions occur, for example, when a point cloud transitions from a single cluster to multiple separated groups. This motivates a model architecture capable of capturing discontinuous, high-order feature interactions. GART 2.0 is implemented using the LightGBM gradient-boosting framework [Ke et al., 2017]. For the task of scalar regression on tabular topological features, Gradient Boosted Decision Trees (GBDTs) offer superior sample efficiency and faster inference compared to deep learning alternatives.

4.1 Model Architecture and Training

We selected LightGBM over standard Random Forests or traditional GBDT implementations due to its "leaf-wise" tree growth strategy. Unlike level-wise algorithms that grow trees depth-first to maintain balance, LightGBM splits the leaf with the maximum delta loss, regardless of depth. This approach is mathematically advantageous for approximating the α ratio because the relationship between graph topology and tour cost often exhibits sharp, non-smooth phase transitions—for example, the sudden drop in α when a random point cloud begins to form distinct clusters. Leaf-wise growth allows the model to allocate more complexity to these critical transition regions without wasting splits on simpler, linear parts of the manifold.

We train the model to minimize the Root Mean Squared Error (RMSE) between the predicted ratio $\hat{\alpha}$ and the ground-truth α, and clip $\hat{\alpha}$ to the theoretical interval $[1.0, 2.0]$ from Equation 4 at inference. In practice, none of the 16,920 predictions on the multi-dimensional test set fall outside this range (observed range $[1.031, 1.948]$), so the clip is a safety net rather than a correction. To optimize the model's generalization capability, we employed the Optuna framework [Akiba et al., 2019] to perform a Bayesian search of the hyperparameter space. The optimization utilized the Tree-structured Parzen Estimator (TPE) algorithm over 100 trials to tune the learning rate, tree structure, and regularization terms. The final model configuration, detailed in Table 4, resulted in a robust ensemble of 2,031 trees. While the hyperparameter search allowed for up to 300 leaves per tree to capture complex interactions, the final trained model exhibited a maximum leaf count of 148 and an average of 148.0 leaves per tree (the Optuna-selected cap), indicating that the constraint was tight enough to prevent unbridled overfitting while allowing the depth needed to capture topological regime transitions.

Table 4: GART 2.0 hyperparameters and trained-model statistics. Tuned block: Optuna TPE, 100 trials, RMSE objective.

Group	**Parameter**	**Value**
Tuned hyperparameters (Optuna TPE, 100 trials, RMSE objective)		
Tree structure	num leaves	148
	min child samples	23
	feature fraction	0.548
	bagging fraction / freq	0.609 / 6
Optim. & reg.	learning rate	0.0260
	L1 regularisation	7.76×10^{-2}
	L2 regularisation	1.19×10^{-5}
Training configuration		
Training setup	boosting type	gbdt
	objective	regression_l2 (RMSE)
	early-stopping patience	100
	random seed	42
Trained-booster statistics		
Booster	trees (after early stop)	2,031
	avg leaves / tree	148.0
	max leaves / tree	148
	avg tree depth	18.89
	max tree depth	33

Inference cost is dominated by MST construction, not tree traversal. The trained model performs $K = 2{,}031$ tree traversals at inference, with an average root-to-leaf depth of $\bar{D} \approx 18.9$ (empirical maximum 33). Since LightGBM grows leaf-wise, depth is not $\lceil \log_2 L \rceil$; the per-prediction cost is $K \cdot \bar{D} \approx 3.8 \times 10^4$ comparisons, independent of n. Feature extraction is dominated by the MST: $O(n^2 \cdot d)$ in general (dense distance matrix), reduced to $O(n \log n)$ in 2D by constructing the MST on a Delaunay-sparse edge set [Delaunay, 1934], since the Euclidean MST is a subgraph of the Delaunay triangulation. Sorting the $n - 1$ MST edges for quantile features adds $O(n \log n)$, and the remaining topological statistics (degrees, diameter via two BFS passes, leaf counts) are $O(n)$. The total asymptotic complexity is therefore $O(n^2 \cdot d)$ in general, and $O(n \log n)$ for 2D inputs. For comparison, the Christofides–Serdyukov $\frac{3}{2}$-approximation requires a Minimum Weight Perfect Matching on the odd-degree MST vertices, with exact algorithms in $O(n^3)$ [Edmonds, 1965, Gabow, 1990, Cook and Rohe, 1999]; approximate MWPM variants reduce this but sacrifice the 1.5 guarantee [Duan and Pettie, 2014, Drake and Hougardy, 2003], and Christofides returns a worst-case upper bound rather than a point estimate.

4.2 Feature importance and design validation

To verify that the 30-feature set is doing real work—rather than carrying redundant descriptors—we compute SHAP values [Lundberg and Lee, 2017] on the held-out test split using the saved booster (see `shap_analyzer_v3.py`). Table 5 ranks all 30 features by mean absolute SHAP contribution. Three observations motivate the final feature set:

1. **MST topology dominates.** The MST dominance ratio alone carries 26.4% of the model's decision mass, and seven of the top ten features are MST-derived (dominance ratio, degree mean, diameter, diameter-normalised, gap ratio, large-edge count, leaf ratio). This is consistent with the BHH intuition that tour length scales with tree-like spanning structure; it also justifies the decision to feature-engineer from the MST rather than from raw coordinates.

2. **Dimension and size are first-class inputs.** n and d together contribute 22.2% of the decision mass, which is how the model adapts to varying problem scale without being given a distribution label.

3. **Lightweight spread statistics add coverage.** Centroid-distance dispersion (4.8% share on `centroid_dist_std` alone, 9.9% summed across the four centroid-distance descriptors) and bounding hypervolume ($\sim$4.6% summed across the log and linear forms) contribute meaningfully once evaluated in the canonical principal-axis frame (§3.1), which is what allows the rotation-sensitive block to carry signal without leaking orientation noise and without requiring a convex-hull computation, keeping the feature set applicable at arbitrary d.

Earlier design iterations carried a larger candidate pool; the reported 30-feature set is the pruned subset retained after SHAP-guided removal of redundant descriptors. We emphasize that SHAP is used here only to audit the *design*; the numerical results reported later are generated by the trained model with no further feature selection.

5 Benchmarking

We evaluate GART 2.0 against established estimators on three benchmarks: a synthetic 2D dataset spanning five distribution classes, a multi-dimensional synthetic dataset covering $d \in \{2, \ldots, 100\}$ (with $d = 100$ reserved as an extrapolation bucket outside the training range), and the 78 EUC_2D instances of TSPLIB95 [Reinelt, 1991]. The 2D suite mixes uniform and non-uniform generators: anisotropic clusters, geometric lattices, and near-linear manifolds. The near-linear generator was not part of the training distribution; we include it explicitly to probe how MST-based estimation handles geometries outside what the model was taught.

Throughout this evaluation, we report **SDPE as the primary metric** (first column in every results table), with MAPE and median absolute percentage error reported alongside for completeness.

The *Standard Deviation of Percentage Error* (SDPE) measures *precision*—the tightness of the error distribution—and is defined as the sample standard deviation (with Bessel's correction, denominator $N-1$) of the signed relative error $e_i = (\hat{L}_i - L_i)/L_i$:

$$\text{SDPE} = 100 \cdot \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(e_i - \bar{e})^2}, \qquad \bar{e} = \frac{1}{N}\sum_{i=1}^{N} e_i.$$

We accompany each SDPE with a 95% bootstrap confidence interval (1000 resamples, random seed 42). The *Mean Absolute Percentage Error* (MAPE) measures accuracy: $\text{MAPE} = \frac{1}{N}\sum |e_i| \times 100$. A precise estimator with a small constant offset can be recalibrated; an imprecise one cannot, so SDPE is the primary quality signal.

5.1 Benchmark Models

We compared GART 2.0 against seven classical baseline estimators representing the evolution of TSP approximation theory, plus GART 1.0 [Feldman, 2025] — an earlier ML-estimator variant that GART 2.0 supersedes. To ensure reproducibility, we present the exact mathematical formulations used in our Python implementation, with constants derived directly from the source code.

Table 5: All 30 features of GART 2.0 ranked by mean |SHAP| (held-out test split, $N = 2{,}000$ sampled rows, seed 42, recomputed on the current trained booster). The bar shows the share of total SHAP mass.

Rank	Feature	Mean |SHAP|	Share (%)
1	mst_dominance_ratio	0.0388	26.43
2	dimension	0.0171	11.66
3	n_customers	0.0155	10.55
4	centroid_dist_std	0.0071	4.81
5	mst_degree_mean	0.0068	4.62
6	mst_diameter	0.0063	4.30
7	mst_diameter_normalized	0.0061	4.18
8	mst_gap_ratio	0.0050	3.39
9	large_edge_count	0.0044	2.98
10	bounding_hypervolume	0.0040	2.70
11	mst_leaf_ratio	0.0039	2.66
12	mst_degree_max	0.0034	2.32
13	centroid_dist_mean	0.0030	2.05
14	log_bounding_hypervolume	0.0027	1.85
15	centroid_dist_max	0.0024	1.61
16	centroid_dist_iqr	0.0022	1.51
17	mst_edge_max	0.0020	1.35
18	mst_edge_q75	0.0017	1.19
19	aspect_ratio	0.0017	1.13
20	mst_degree_std	0.0016	1.11
21	log_node_density	0.0015	1.01
22	mst_edge_q90	0.0014	0.97
23	mst_edge_q25	0.0014	0.93
24	mst_edge_mean	0.0012	0.81
25	node_density	0.0011	0.76
26	mst_edge_skew	0.0011	0.72
27	mst_edge_q10	0.0010	0.70
28	mst_edge_q50	0.0010	0.68
29	mst_edge_kurtosis	0.0009	0.59
30	mst_edge_std	0.0006	0.41

5.1.1 GART 1.0 (prior work)

Feldman [2025] introduced the original Geometrically Assisted Regression Tree (GART 1.0) as a gradient-boosted estimator trained on a 2D-specific feature set combining convex hull area, PCA axes, and MST edge statistics. GART 1.0 is included as the direct ML-estimator predecessor to the model presented here. GART 1.0 originated as an unpublished technical report; we include it here as an ablation against the same features the present model replaces, not as a peer-reviewed benchmark. Because GART 1.0's feature extractor relies on 2D-only geometric primitives (Graham-scan hull, 2D PCA), it is reported on the 2D synthetic and TSPLIB95 EUC_2D benchmarks but is not applicable in the $d \geq 3$ setting. Its scores in Tables 9 and 10 quantify the accuracy gap closed by GART 2.0's MST-centric redesign.

5.1.2 Beardwood-Halton-Hammersley (BHH) Theorem

The foundational asymptotic bound for TSP assumes the tour cost scales with the hypervolume of the vertex set [Beardwood et al., 1959]. Our implementation estimates the tour cost L as:

$$\hat{L}_{\text{BHH}} = \beta_d \cdot n^{\frac{d-1}{d}} \cdot V^{\frac{1}{d}} \tag{5}$$

where V is the volume of the convex hull (or bounding box in high dimensions) and β_d is a dimension-dependent constant. For $d = 2$ we use the large-scale empirical estimate $\beta_2 = 0.7124 \pm 0.0002$ of Johnson et al. [1996], obtained from Concorde-verified optima on uniform random point sets and heuristic extensions to larger n. An independent statistical-mechanics study [Percus and Martin, 1996] reports $\beta_2 = 0.7120 \pm 0.0002$ under toroidal boundary conditions; the two estimates differ by roughly 2σ, consistent with the expected open-vs-toroidal boundary correction. We retain 0.7124 because the benchmark BHH implementation consumes uniform point sets under open (square) boundaries, matching the calibration setting of Johnson et al. [1996] more tightly than the toroidal setting of Percus and Martin [1996]. A separate statistical-mechanics analysis by Cerf et al. [1997] reports $\beta_3 = 0.6979 \pm 0.0002$ for the 3D case; we do not apply BHH for $d \geq 3$ because the hull-to-bounding-box volume ratio decays exponentially with dimension, making V unstable as an estimator [Carlsson and Yu, 2024].

5.1.3 Cavdar–Sokol (CS) Distribution-Free Estimator

To handle anisotropic distributions, Cavdar and Sokol [2015] proposed a distribution-free estimator that replaces simple volume metrics with statistical moments of coordinate deviations. The estimator is defined in 2D:

$$\hat{L}_{\text{CS}} = \alpha_1 \sqrt{n \cdot \sigma'_x \sigma'_y} + \alpha_2 \sqrt{\frac{n \cdot A \cdot \sigma_x \sigma_y}{\bar{c}_x \bar{c}_y}} \tag{6}$$

where σ_k is the standard deviation of coordinates in dimension k, $\bar{c}_k$ is the average absolute distance from each node to the *midpoint axis* of the bounding rectangle in dimension k (that is, to the line $x_k = (\max_k + \min_k)/2$, not to the coordinate mean), σ'_k is the standard deviation of those absolute distances, and A is the convex-hull area of the node set. The regression coefficients are fitted by Cavdar and Sokol [2015] as $\alpha_1 = 2.791$ and $\alpha_2 = 0.2669$. Three details of the implementation deserve emphasis, because earlier public adaptations of the formula silently departed from each one and produced inflated error. First, the paper is explicit that the dispersion ratio in the second term uses distances to the rectangle's midpoint axes, not to the centroid; we follow the paper exactly. Second, the paper states that on tests over non-random graphs "we used the area of the convex hull of the nodes as A", which we implement with `scipy.spatial.ConvexHull` (falling back to the axis-aligned bounding-box volume only when the hull is degenerate, i.e. $n \leq d + 1$ or colinear points). Third, Cavdar–Sokol trained on axis-aligned rectangular graphs, so applying the estimator to an arbitrarily oriented point cloud without realignment would distort every per-axis statistic; we therefore rotate coordinates into the minimum-area bounding rectangle using rotating calipers over the convex hull [Freeman and Shapira, 1975], after which all dispersion statistics and the convex-hull area are computed in that canonical frame. For $n < 1000$ we additionally apply the small-n correction of Cavdar and Sokol [2015, Eq. 4], $E/T = 0.9325\, e^{0.00005298\, n} - 0.2972\, e^{-0.01452\, n}$, by dividing the raw estimate by the factor; products of the per-axis statistics are evaluated in log-space so the formula remains finite under our d-dimensional generalisation, where the paper's 2D form is lifted by the geometric-mean convention $\text{term}_1 = \alpha_1 n^{(d-1)/d} (\prod_j \sigma'_j)^{1/d}$ and analogously for the second term, reducing exactly to Eq. (6) when $d = 2$.

5.1.4 Kwon–Golden–Wasil (1995) Estimator

Kwon et al. [1995] proposed a regression-based TSP estimator calibrated on depot-centric VRP instances. It is the only classical baseline in our comparison that is explicitly sensitive to both n and a characteristic radius R of the point set:

$$\hat{L}_{\text{Kwon}} = \left(0.8326 - 0.0011\, n + 1.1147\, \frac{R}{n}\right) \sqrt{n\, A}, \tag{7}$$

where A is the convex hull area and R is the mean distance from each vertex to the centroid (a depot-free adaptation of Kwon's original depot-to-region distance, matching the specification used by Cavdar and Sokol [2015] when they benchmarked Kwon). Two properties of this form are worth noting. First, the linear $-0.0011\, n$ term was calibrated for VRP customer counts in the tens-to-hundreds range; extrapolated to n in the thousands or tens of thousands it dominates the bracket and drives the prediction negative, which we clip to zero. Second, to apply the formula across

TSPLIB's wide range of raw coordinate scales, we rescale coordinates so the bounding-box diagonal is 1, evaluate Kwon in that normalized space, and multiply the result by the original diagonal; without this step the R term becomes scale-dependent and unusable across TSPLIB. Both behaviours are visible directly in Table 10: Kwon tracks the data well for $n \leq 400$ and degenerates for $n \geq 1500$.

5.1.5 Daganzo (1984) Estimator

Daganzo [1984a] derived a square-root CVRP approximation—structurally a BHH-form estimator with a calibrated constant—for uniformly distributed points on a rectangular service region:

$$\hat{L}_{\text{Daganzo}} = 0.57\,\sqrt{n\,A}. \tag{8}$$

Structurally this is BHH with a different constant: the 0.57 coefficient is the CVRP calibration reported in Daganzo [1984a] (the companion paper Daganzo [1984b] treats TSP tours in zones of different shapes separately), visibly looser than the BHH/uniform-square value 0.7124 we use for the BHH baseline. We include both separately because the literature reports them as distinct benchmarks; Table 10 shows Daganzo is competitive with BHH at large n (where both track the BHH asymptote) and weaker at small n where the constant-k assumption misses instance-specific structure.

5.1.6 Chien Estimator

Chien [1992] proposed a probabilistic estimator for logistics route planning that scales tour cost with the square root of the number of stops and the service area:

$$\hat{L}_{\text{Chien}} = k_1\sqrt{n \cdot A} + k_2\frac{n}{p} \tag{9}$$

where A is the convex hull area, n is the number of stops, p is the number of routes, and k_1, k_2 are empirically fitted constants ($k_1 = 0.98$, $k_2 = 0$, $p = 1$ for a single tour). Like BHH, this estimator depends on convex hull area and is restricted to 2D.

5.1.7 Hilbert Space-Filling Curve

This constructive heuristic maps d-dimensional points onto a 1-dimensional interval $[0, 1]$ using a fractal space-filling curve; the original heuristic of Bartholdi and Platzman [1982] used a Sierpinski-style curve, and the Hilbert-curve variant we implement here is a later refinement in the same family. The points are sorted by their Hilbert index $h(x)$, and the tour is constructed by visiting vertices in this sorted order:

$$\hat{L}_{\text{Hilbert}} = \sum_{i=1}^{n-1} \|x_{(i)} - x_{(i+1)}\|_2 + \|x_{(n)} - x_{(1)}\|_2 \tag{10}$$

This approach preserves locality without constructing a full distance matrix, offering $O(n \log n)$ complexity.

5.1.8 MST Ratio

The control baseline scales the weight of the Minimum Spanning Tree by a dimension-dependent factor $\rho(d)$ derived from empirical observation of the MST-to-TSP ratio [Percus and Martin, 1996]:

$$\hat{L}_{\text{MST-Ratio}} = \rho(d) \cdot L_{\text{MST}} \tag{11}$$

For 2D instances, we use $\rho(2) = 1.075$, computed as $\beta_{\text{TSP}}(2)/\beta_{\text{MST}}(2)$ using $\beta_{\text{TSP}}(2) \approx 0.7124$ [Johnson et al., 1996] and the empirical $\beta_{\text{MST}}(2) \approx 0.663$ from Cortina-Borja and Robinson [2000] (consistent with the Avram and Bertsimas [1992] analytic lower bound of 0.6008). For higher dimensions, we use $\rho(3) = 1.05$ and $\rho(d) = 1.0 + 0.075 \cdot (2/d)$ for $d \geq 4$, reflecting the known convergence of the MST-to-TSP ratio as dimensionality increases [Percus and Martin, 1996]. These constants were calibrated for uniform distributions; the elevated SDPE observed for MST Ratio in Table 8 reflects the inability of a fixed constant to adapt to the topological variation present in non-uniform test instances.

5.2 Datasets

We evaluate on three datasets: a multi-dimensional synthetic set, a 2D diverse synthetic set, and the EUC_2D subset of TSPLIB95. The synthetic sets are held-out test splits from our generation pipeline (Appendix B); TSPLIB95 is real-world. The evaluation pipeline assumes symmetric metric distances. Because $L_{\text{TSP}} \leq 2\, L_{\text{MST}}$ holds for any metric TSP (double-tree bound), we drop from every aggregate any instance for which $L_{\text{TSP}}/L_{\text{MST}} > 2.5$—a deterministic, model-agnostic filter that absorbs integer-rounded EXPLICIT matrices while flagging instances mathematically inconsistent with the metric assumption. Under this rule the only TSPLIB95 instance currently excluded is the bridge graph `brg180`, whose published optimum satisfies $L_{\text{TSP}}/L_{\text{MST}} \approx 160$.

5.2.1 Multi-dimensional synthetic ($d = 2$–100)

16,920 held-out instances with $n \in [5, 1000]$ and $d \in \{2, 3, \ldots, 50, 100\}$, sampled by stratified dimension–size–grid buckets across 16 per-dimension distribution types drawn from the generators used for our training set: uniform, normal, clustered (k-means [Lloyd, 1982]), and autoregressive-correlated. Stratification enforces zero overlap with the training split. $d = 100$ is outside the training range and serves as the extrapolation test. The full (d, n) count grid is given in Appendix B.1 (Table 12).

5.2.2 2D diverse synthetic

2,580 instances, $d = 2$, $n \in [5, 1000]$, coordinate grid side-length $G \in [1000, 10000]$. Five distribution classes (Table 6) each probe a different assumption made by classical 2D estimators. Four classes (isotropic, biased, geometric, clustered) follow conventions used in the TSP-instance-generation literature, including the variants reported by Cavdar and Sokol [2015] for the distribution-free baseline. The fifth class, *Line Noise*, generates near-linear manifolds of the form $y = mx + b + \epsilon$ with small ϵ. Near-collinear point sets are a standard degenerate TSP test case [cf. Papadimitriou, 1977]; we include this class precisely to measure how the model handles point sets that collapse toward a 1D manifold, where hull-based formulas vanish and bounding-box formulas over-predict. This geometry is *not* represented in the training distribution.

Table 6: 2D benchmark dataset composition (2,580 instances).

Class	Generators	Count	Stress target
Isotropic [Cavdar and Sokol, 2015, Golden and Stewart, 1985]	Uniform, Normal, Triangular, Truncated Exp.	840	Control: $A_{\text{hull}} \approx A_{\text{box}}$.
Biased / Skewed [Cavdar and Sokol, 2015]	Squeezed, Uni-Tri, Tri-Squeezed, Correlated	840	Aspect-ratio distortion: $\sigma_x \neq \sigma_y$.
Geometric Struct. [Reinelt, 1991, Bossek et al., 2019]	Grid, Boundary (hollow), X-Central	630	Lattice regularity, hollow voids.
Clustered [Golden and Stewart, 1985, Bossek et al., 2019]	k-Means ($k \in [5, 20]$)	60	Bimodal edge-weight distribution.
Line Noise†	Linear manifold $y = mx + b + \epsilon$	210	Hull/box divergence as data collapses toward 1D.
Total		**2,580**	

† Not represented in training; included to measure generalization to unseen geometries.

5.2.3 TSPLIB95 EUC_2D common subset

78 EUC_2D instances with n from 14 to 85,900 [Reinelt, 1991]. This is the intersection where every benchmark model is directly applicable: all classical baselines require 2D Euclidean coordinates, and the MST Ratio baseline needs only a distance matrix. The remaining 32 metric TSPLIB95 instances (CEIL_2D, ATT pseudo-Euclidean, GEO great-circle, EXPLICIT-metric) are not natively Euclidean and are evaluated in Section 6 via MDS embedding, which only GART 2.0 supports. For large instances where the dense $n \times n$ distance matrix would exceed memory, we replace it with a Delaunay-based MST construction [Delaunay, 1934] that uses only $O(n)$ edges.

5.3 Results

Every table below reports six quantities per model row: SDPE with a 95% bootstrap confidence interval (1,000 resamples, seed 42) as the primary precision metric, MAPE, median $|\%\,\text{error}|$, R^2 on tour cost, R^2_α on the MST-ratio $\alpha = L_{\text{TSP}}/L_{\text{MST}}$ (coefficient of determination of $\hat{\alpha}$ against the ground-truth α), and median prediction time in milliseconds (feature extraction + MST + inference). The per-bucket instance count is listed once in the bucket-label cell rather than repeated per row. We introduce R^2_α as a complement to R^2: on high-density or high-dimensional instances the MST length alone explains most of the tour cost variance, so R^2 saturates near 1.0 for any MST-based estimator. R^2_α measures how well a model predicts the multiplicative correction on top of the MST baseline and is therefore the discriminative signal for comparing MST-based estimators; it is defined only for models that produce a learned $\hat{\alpha}$ (GART 2.0, GART 1.0) and reported as "—" elsewhere. Within each bucket, model rows are ascending so the strongest estimator on the precision axis appears first.

Each bucket closes with a separate "*Optimal solver*" row whose only populated cell is the Time column. We define the **optimal solver time** as the shorter of the two wall-clock times from locally running CONCORDE [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017] on a single core, without any timeout, where both solvers report the same tour length (and that tour length matches the published TSPLIB optimum where applicable). When only one solver reaches the verified optimum, that solver's time is used; when neither is locally tractable (very large n), the bucket is annotated with a citation to published benchmark times. The column reports milliseconds in scientific notation so the six orders of magnitude spanning small-n and large-n instances remain legible and directly comparable to the model times in the same column.

5.3.1 Multi-dimensional benchmark

We present the 16,920-instance multi-dimensional benchmark in two orthogonal slices: by instance size (Table 7) and by dimension (Table 8). Both tables summarize the same underlying data.

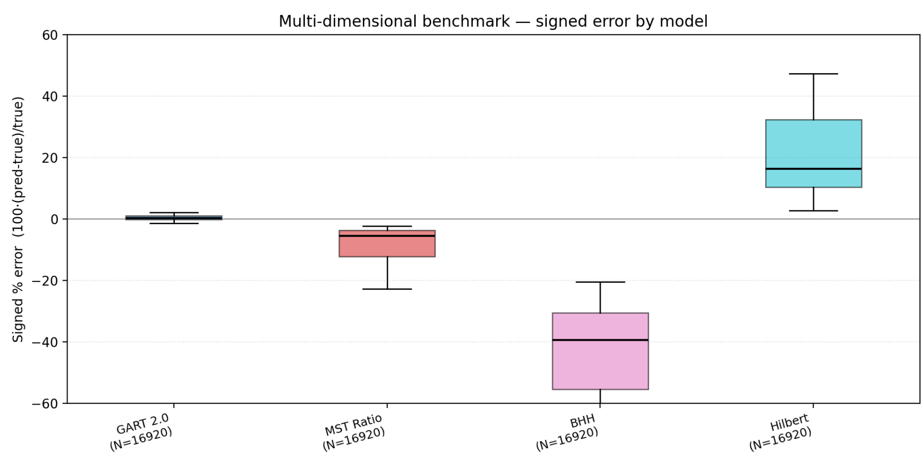

Figure 1: Percent-error distribution per model on the ND benchmark (16,920 instances).

GART 2.0 reaches 0.88% MAPE and 1.28% SDPE overall, against 8.54% / 6.84% for MST Ratio and 21.20% / 14.38% for Hilbert. Precision tightens monotonically as n grows (SDPE drops from 1.98% at $n \leq 10$ to 0.55% at $n \in [501, 1000]$) and as d grows (2.97% at $d = 2$ down to 0.61% at $d \in [30, 50]$), consistent with the theoretical convergence $\alpha \to 1$ in high dimensions [Percus and Martin, 1996]. The $d = 100$ bucket sits outside the training grid (which caps at $d = 50$) and therefore serves as a transfer test: the feature set must remain predictive at a dimension never seen during training. We acknowledge this is extrapolation in the *favorable* direction—α concentrates as d grows, so the target variance shrinks with dimension—so the claim here is feature-set transferability, not that $d = 100$ is

Table 7: ND benchmark by instance-size bucket.

Bucket	Model	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	R^2	R^2_α	Time (ms)
$[5, 10]$ Count=4,230	**GART 2.0**	**1.98** [1.90, 2.06]	1.33	0.89	1.000	0.935	116
	MST Ratio	5.31 [5.15, 5.47]	18.80	17.92	0.966	—	4.53
	Hilbert	7.49 [7.17, 7.82]	8.15	5.89	0.995	—	0.66
	Optimal solver	—	—	—	—	—	1 ms
$[11, 50]$ Count=2,820	**GART 2.0**	**1.35** [1.28, 1.42]	1.06	0.90	1.000	0.906	114
	MST Ratio	2.99 [2.85, 3.13]	7.46	6.92	0.996	—	26
	Hilbert	12.41 [11.98, 12.89]	19.59	15.81	0.981	—	11
	Optimal solver	—	—	—	—	—	108 ms
$[51, 100]$ Count=3,525	**GART 2.0**	**1.03** [1.00, 1.05]	0.86	0.61	1.000	0.916	131
	MST Ratio	2.20 [2.12, 2.28]	5.36	4.77	0.998	—	42
	Hilbert	13.54 [13.19, 13.92]	24.12	20.15	0.973	—	32
	Optimal solver	—	—	—	—	—	76 ms
$[101, 200]$ Count=705	**GART 2.0**	**0.61** [0.58, 0.65]	0.55	0.50	1.000	0.954	147
	MST Ratio	1.66 [1.57, 1.77]	4.36	4.00	0.999	—	81
	Hilbert	13.12 [12.62, 13.65]	26.84	23.82	0.965	—	73
	Optimal solver	—	—	—	—	—	224 ms
$[201, 500]$ Count=2,115	**GART 2.0**	**0.55** [0.54, 0.56]	0.51	0.37	1.000	0.953	176
	MST Ratio	1.46 [1.42, 1.49]	4.03	3.82	0.999	—	129
	Hilbert	12.63 [12.40, 12.86]	28.18	26.23	0.960	—	144
	Optimal solver	—	—	—	—	—	886 ms
$[501, 1000]$ Count=3,525	**GART 2.0**	**0.55** [0.54, 0.57]	0.48	0.29	1.000	0.949	272
	MST Ratio	1.33 [1.30, 1.35]	3.82	3.64	0.999	—	334
	Hilbert	12.72 [12.57, 12.88]	29.91	28.50	0.953	—	825
	Optimal solver	—	—	—	—	—	3,799 ms
Total Count=16,920	**GART 2.0**	**1.28** [1.24, 1.31]	0.88	0.65	1.000	0.974	171
	MST Ratio	6.84 [6.75, 6.94]	8.54	5.58	0.999	—	47
	Hilbert	14.38 [14.24, 14.52]	21.20	16.21	0.962	—	52
	Optimal solver	—	—	—	—	—	122 ms

intrinsically harder than the training range. SDPE rises modestly from 0.61% at $d \in [30, 50]$ to 0.65% at $d = 100$, while MAPE moves from 0.38% to 1.04%. The two moves together describe a distribution-shift bias rather than a variance blow-up: the error distribution stays tight (SDPE tracks the shrinking target variance) but its center shifts slightly under genuine extrapolation, which is the benign failure mode for a precision-first estimator. The transfer gap is bounded and monotone in the metric that matters for the precision-first framing. MST Ratio's SDPE is nearly flat because its fixed constant does not adapt to instance structure. Hilbert degrades with n because space-filling locality breaks on non-uniform distributions; this applies everywhere $d = 100$ appears in this section.

5.3.2 2D diverse benchmark

The 2D diverse benchmark pushes every estimator against five distribution classes that probe distinct failure modes of classical geometry-based estimators. Across the 2,580 instances GART 2.0 holds SDPE under 6.2% in every size bucket and under 3.3% for instances with $n \geq 500$, outranking the MST-ratio baseline by roughly a factor of two on the overall aggregate. Table 9 disaggregates the results by instance size.

Table 8: ND benchmark by dimension; same 16,920 instances as Table 7.

Bucket	Model	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	R^2	R^2_α	Time (ms)
$d = 2$ Count=648	**GART 2.0**	**2.97** [2.72, 3.21]	1.98	1.18	1.000	0.947	144
	MST Ratio	10.51 [9.88, 11.10]	13.09	8.88	0.995	—	45
	Hilbert	15.78 [14.81, 16.79]	29.71	35.19	0.805	—	1.25
	Optimal solver	—	—	—	—	—	102 ms
$d = 3$–5 Count=1,944	**GART 2.0**	**2.03** [1.91, 2.14]	1.30	0.69	1.000	0.951	160
	MST Ratio	7.91 [7.61, 8.18]	11.64	8.24	0.996	—	47
	Hilbert	15.72 [15.24, 16.20]	35.70	40.13	0.732	—	1.62
	Optimal solver	—	—	—	—	—	109 ms
$d = 6$–10 Count=3,240	**GART 2.0**	**1.28** [1.22, 1.34]	0.82	0.45	1.000	0.970	167
	MST Ratio	6.56 [6.36, 6.74]	10.21	7.15	0.997	—	42
	Hilbert	13.83 [13.52, 14.12]	32.53	35.93	0.787	—	10.8
	Optimal solver	—	—	—	—	—	122 ms
$d = 15$–25 Count=1,944	**GART 2.0**	**0.85** [0.80, 0.90]	0.54	0.30	1.000	0.984	176
	MST Ratio	6.02 [5.80, 6.23]	8.68	5.63	0.998	—	45
	Hilbert	11.73 [11.46, 12.00]	24.60	26.36	0.844	—	41
	Optimal solver	—	—	—	—	—	123 ms
$d = 30$–50 Count=3,240	**GART 2.0**	**0.61** [0.58, 0.64]	0.38	0.21	1.000	0.991	175
	MST Ratio	5.96 [5.80, 6.13]	7.65	4.59	0.999	—	47
	Hilbert	8.00 [7.86, 8.15]	17.11	18.19	0.925	—	55
	Optimal solver	—	—	—	—	—	136 ms
$d = 100$ Count=5,904	**GART 2.0**	**0.65** [0.64, 0.67]	1.04	0.97	1.000	0.973	178
	Hilbert	4.60 [4.54, 4.66]	10.40	11.07	0.971	—	99
	MST Ratio	5.93 [5.81, 6.04]	6.55	3.44	0.999	—	50
	Optimal solver	—	—	—	—	—	120 ms
Total Count=16,920	**GART 2.0**	**1.28** [1.24, 1.31]	0.88	0.65	1.000	0.974	171
	MST Ratio	6.84 [6.75, 6.94]	8.54	5.58	0.999	—	47
	Hilbert	14.38 [14.24, 14.52]	21.20	16.21	0.962	—	52
	Optimal solver	—	—	—	—	—	122 ms

GART 2.0 leads at 3.33% MAPE / 5.19% SDPE overall, with GART 1.0 [Feldman, 2025] second among ML estimators at 7.35% / 7.95% — the relative improvement over the predecessor model is most pronounced at $n \geq 200$, where GART 2.0's MST-centric features generalize to larger instances that GART 1.0's hull-based features cannot track. Among classical baselines, MST Ratio is strongest at 12.45% / 11.06%. Cavdar's MAPE drops by roughly an order of magnitude, from $\sim$160% in earlier public adaptations of the formula to 16.13% here, with SDPE tightening in parallel from $> 200\%$ to 32.19%; the remaining spread still exceeds Kwon and Daganzo on this benchmark. The gain is traceable to the implementation corrections discussed in §5.1.8 (rotation into the minimum-area bounding rectangle before evaluating per-axis statistics; the midpoint-axis distance in $\bar{c}_k$ rather than the centroid distance; the convex-hull area as A per the paper's own Section 5). BHH, Chien, and Daganzo all exceed 25% SDPE because their enclosure-based formulas amplify variance on the Biased and Line Noise classes, where bounding-box and hull volumes diverge. Hilbert has intermediate MAPE (31%) but lower SDPE (14%): its errors are large but consistent — a systematic space-filling bias. Kwon (1995) is structurally invalid above $n \approx 750$ (its linear $-0.0011\,n$ term forces the estimate negative) and is reported only where applicable.

Table 9: 2D diverse synthetic benchmark.

Bucket	Model	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	R^2	R^2_α	Time (ms)
$[5, 10]$ Count=360	**GART 2.0**	**5.88** [5.46, 6.27]	4.69	3.73	0.991	0.830	64
	GART 1.0	6.93 [6.27, 7.52]	5.27	4.09	0.984	0.784	31
	Daganzo	7.79 [7.25, 8.28]	67.45	66.16	-0.028	—	3.85
	Hilbert	8.84 [8.11, 9.50]	8.35	5.79	0.965	—	0.18
	BHH	9.74 [9.05, 10.36]	59.32	57.70	0.203	—	3.87
	MST Ratio	9.99 [9.33, 10.65]	28.95	29.17	0.785	—	8.77
	Kwon	11.28 [10.42, 12.06]	50.16	48.60	0.422	—	4.61
	Chien	13.40 [12.47, 14.27]	44.04	41.81	0.545	—	3.82
	Cavdar	20.11 [18.79, 21.35]	26.54	23.71	0.756	—	8.43
	Optimal solver	—	—	—	—	—	82 ms
$[11, 50]$ Count=480	**GART 2.0**	**6.29** [5.72, 6.81]	4.49	2.99	0.993	0.679	59
	GART 1.0	6.43 [5.85, 6.96]	4.56	2.94	0.991	0.715	37
	Daganzo	7.43 [6.92, 7.96]	42.70	42.10	0.592	—	4.50
	BHH	9.28 [8.63, 9.93]	28.38	27.64	0.817	—	4.38
	Kwon	9.55 [8.76, 10.29]	19.45	18.38	0.908	—	4.72
	MST Ratio	9.66 [8.86, 10.43]	15.16	13.63	0.949	—	12
	Hilbert	10.37 [9.72, 11.07]	27.18	26.95	0.804	—	0.35
	Cavdar	11.27 [10.26, 12.18]	9.88	9.10	0.972	—	10
	Chien	12.77 [11.89, 13.69]	10.03	8.51	0.967	—	4.51
	Optimal solver	—	—	—	—	—	85 ms
$[51, 100]$ Count=630	**GART 2.0**	**5.91** [5.32, 6.50]	3.70	2.09	0.995	0.597	80
	GART 1.0	7.38 [6.79, 8.01]	5.39	3.52	0.987	0.494	43
	MST Ratio	8.59 [7.82, 9.36]	11.45	9.34	0.975	—	16
	Hilbert	9.82 [9.11, 10.53]	34.23	34.08	0.734	—	0.60
	Daganzo	10.47 [9.56, 11.32]	30.37	31.50	0.735	—	5.28
	Kwon	13.00 [11.93, 14.09]	13.14	11.59	0.936	—	5.59
	BHH	13.08 [11.97, 14.20]	16.22	14.98	0.912	—	5.16
	Chien	18.00 [16.36, 19.63]	22.06	17.77	0.887	—	5.13
	Cavdar	19.50 [17.53, 21.43]	13.59	7.98	0.945	—	11
	Optimal solver	—	—	—	—	—	248 ms
$[101, 500]$ Count=500	**GART 2.0**	**4.48** [3.87, 5.03]	2.57	1.22	0.997	0.535	68
	GART 1.0	5.83 [5.24, 6.45]	9.41	8.51	0.982	-0.903	75
	MST Ratio	6.31 [5.53, 6.96]	7.46	6.06	0.991	—	27
	Hilbert	10.10 [8.94, 11.23]	38.11	38.03	0.722	—	15
	Kwon†	19.45 [15.66, 22.74]	27.16	27.51	0.775	—	6.42
	Daganzo	23.93 [20.65, 26.97]	25.59	22.54	0.833	—	5.59
	BHH	29.91 [25.40, 33.86]	14.50	4.70	0.931	—	5.43
	Chien	41.15 [34.93, 46.43]	45.10	35.06	0.674	—	5.31
	Cavdar	38.60 [33.11, 43.50]	17.61	3.78	0.923	—	17
	Optimal solver	—	—	—	—	—	2,965 ms
$[501, 1000]$ Count=610	**GART 2.0**	**3.05** [2.73, 3.35]	1.88	0.95	0.997	0.571	100
	MST Ratio	4.54 [4.07, 4.95]	5.71	5.00	0.994	—	45
	GART 1.0	6.38 [5.75, 6.99]	11.09	9.99	0.984	-5.820	108
	Hilbert	11.17 [10.05, 12.30]	38.24	38.21	0.728	—	51
	Daganzo	26.92 [22.92, 30.86]	23.36	17.99	0.850	—	8.05
	BHH	33.64 [28.77, 38.39]	14.75	5.08	0.931	—	8.18
	Chien	46.28 [39.18, 52.99]	49.92	42.71	0.610	—	8.15
	Cavdar	41.94 [35.31, 48.05]	16.34	3.32	0.922	—	23
	Optimal solver	—	—	—	—	—	9,032 ms
Total Count=2,580	**GART 2.0**	**5.19** [4.94, 5.42]	3.33	1.78	0.998	0.840	77
	GART 1.0	7.95 [7.65, 8.22]	7.35	6.27	0.988	0.634	50
	MST Ratio	11.06 [10.70, 11.41]	12.45	8.16	0.992	—	19
	Hilbert	14.24 [13.79, 14.66]	31.01	34.99	0.801	—	0.73
	Kwon†	20.10 [19.15, 20.87]	24.67	18.97	0.872	—	5.24
	Daganzo	25.64 [24.10, 27.16]	35.25	31.49	0.871	—	5.49
	BHH	32.05 [30.14, 33.99]	23.82	15.55	0.943	—	5.45
	Cavdar	32.19 [29.52, 34.85]	16.13	7.09	0.944	—	13
	Chien	44.08 [41.11, 46.66]	33.94	31.21	0.745	—	5.43
	Optimal solver	—	—	—	—	—	459 ms

† Kwon's $-0.0011\,n$ term drives the estimate negative for $n \geq 750$ and is clipped to zero; rows past this threshold are excluded from the Kwon aggregate.

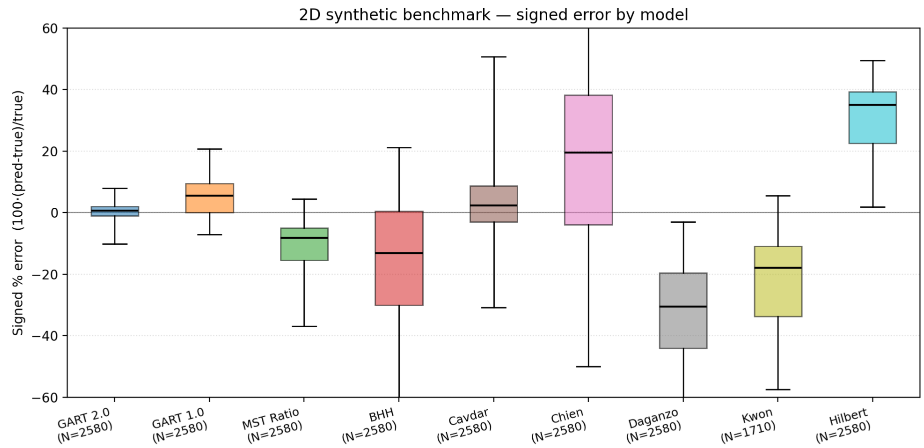

Figure 2: Percent-error distribution per model on the 2D diverse benchmark (2,580 instances).

5.3.3 TSPLIB95 EUC_2D benchmark

Table 10 reports per-bucket results across all 78 EUC_2D instances, with n spanning 51 (`eil51`) to 18,512 (`d18512`). Bucket boundaries (23/16/39 instances) separate small, medium, and large real-world instances; the final $n > 400$ bucket aggregates the asymptotic regime where classical MST-ratio calibration approaches its limit.

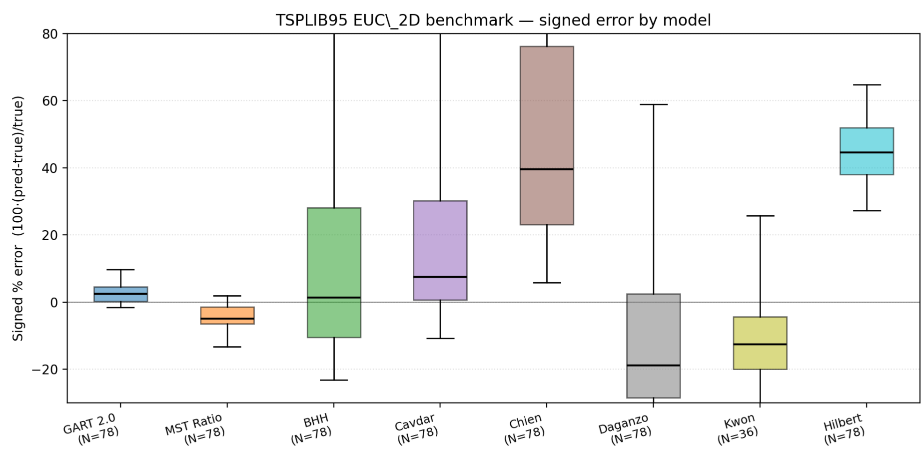

Figure 3: Percent-error distribution per model on the TSPLIB95 EUC_2D subset (78 instances).

Real-world transfer is summarized in Table 10: GART 2.0 retains the tightest SDPE in every bucket, though the margin collapses to 0.43 pp (3.42% vs. 3.85%) in the $n > 400$ bucket where MST Ratio's fixed constant matches the asymptotic α concentration and the two bootstrap intervals overlap. Overall GART 2.0 leads at SDPE 3.42% / MAPE 3.27% versus SDPE 4.54% / MAPE 5.22% for MST Ratio, the only classical that stays competitive. GART 1.0 [Feldman, 2025], the prior 2D estimator, places second among ML models at SDPE 7.44% / MAPE 8.46%; its convex-hull-dependent features degrade on the largest instances, where GART 2.0's MST-centric redesign holds steady. Classical baselines BHH, Cavdar, Chien, and Daganzo degrade past $n = 400$ as real-world instances deviate from the uniform-density assumptions their derivations require. The largest instance in this subset, `d18512` ($n = 18{,}512$), is more than $18\times$ larger than any training instance and is predicted in under half a second. All 78 EUC_2D instances run end-to-end for every model in this table; earlier releases gated academic baselines at $n = 5000$ to bound the $O(n \log n)$ hull-and-calipers preparation cost, but that gate is no longer active because the classical baselines complete in tens

Table 10: TSPLIB95 EUC_2D benchmark. Time is median wall clock.

n range	Model	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	R^2	R^2_α	Time (ms)
$n \in [51, 150]$ Count=23	**GART 2.0**	**3.29** [1.76, 4.22]	2.67	1.57	0.996	0.277	9.68
	MST Ratio	4.05 [2.70, 4.84]	6.93	6.12	0.978	—	1.64
	GART 1.0	5.64 [3.58, 7.19]	5.44	3.92	0.988	-1.307	5.29
	Hilbert	11.46 [8.20, 13.91]	39.91	38.92	0.758	—	1.02
	Daganzo	12.80 [6.16, 16.72]	27.94	29.61	0.894	—	0.68
	Cavdar	13.91 [7.08, 19.14]	9.59	6.71	0.944	—	1.71
	Kwon	14.27 [6.21, 18.53]	14.83	14.01	0.965	—	0.78
	BHH	15.99 [7.69, 20.89]	16.09	13.59	0.958	—	0.96
	Chien	22.00 [10.58, 28.74]	25.66	21.02	0.700	—	0.68
	Optimal solver	—	—	—	—	—	854 ms
$n \in [151, 400]$ Count=16	**GART 2.0**	**3.46** [2.17, 4.40]	2.96	2.69	0.995	0.516	10
	MST Ratio	5.25 [2.45, 7.17]	5.94	4.93	0.986	—	2.54
	GART 1.0	5.52 [3.87, 6.54]	6.03	5.31	0.985	-0.486	6.31
	Hilbert	7.27 [3.45, 10.25]	44.39	42.42	0.596	—	2.18
	Daganzo	18.63 [9.26, 23.67]	21.33	23.29	0.908	—	0.72
	Kwon	22.77 [13.36, 26.77]	22.41	25.27	0.895	—	0.87
	BHH	23.29 [11.57, 29.59]	16.06	6.53	0.793	—	1.00
	Cavdar	27.59 [11.14, 34.47]	20.96	11.34	0.560	—	1.86
	Chien	32.04 [15.91, 40.70]	48.02	36.86	-0.153	—	0.74
	Optimal solver	—	—	—	—	—	15,525 ms
$n > 400$ Count=39	**GART 2.0**	**3.42** [2.68, 3.92]	3.75	2.57	1.000	-0.549	21
	MST Ratio	3.85 [3.06, 4.49]	3.92	3.19	0.998	—	12
	GART 1.0	7.64 [6.07, 8.92]	11.23	9.66	0.978	-10.726	23
	Hilbert	12.55 [9.60, 15.71]	48.66	47.03	0.614	—	12
	Daganzo	39.33 [22.24, 51.03]	25.62	15.21	0.994	—	1.38
	Cavdar	44.70 [26.17, 59.49]	33.49	19.61	0.889	—	2.72
	BHH	49.16 [27.80, 63.78]	34.24	15.53	0.877	—	1.81
	Chien	67.62 [38.24, 87.73]	78.22	57.64	0.249	—	1.29
	Optimal solver	—	—	—	—	—	$>$ 1 h
Total Count=78	**GART 2.0**	**3.42** [2.87, 3.89]	3.27	2.52	1.000	0.208	12
	MST Ratio	4.54 [3.58, 5.37]	5.22	4.96	0.998	—	3.61
	GART 1.0	7.44 [6.23, 8.45]	8.46	6.51	0.978	-3.564	8.47
	Hilbert	11.84 [9.83, 13.59]	45.20	44.60	0.623	—	3.59
	Kwon†	17.49 [12.84, 20.85]	17.57	16.73	0.939	—	0.79
	Daganzo	32.60 [21.00, 42.89]	25.42	23.68	0.994	—	0.94
	Cavdar	36.13 [24.08, 48.14]	23.87	11.07	0.891	—	1.96
	BHH	40.75 [26.25, 53.61]	25.16	13.41	0.880	—	1.13
	Chien	56.05 [36.11, 73.74]	56.53	39.51	0.267	—	0.82
	Optimal solver	—	—	—	—	—	$>$ 1 h

† Kwon (1995) was calibrated for VRP customer counts $\approx$ 50–300; its linear $-0.0011n$ term drives the estimate negative at large n, where it is clipped to 0 and excluded from this aggregate.

of milliseconds even on the largest TSPLIB instance (`pla85900`, $n = 85{,}900$: BHH 50 ms, Cavdar 65 ms, Daganzo 90 ms), with the total-time median for that instance across all models still well under a second.

5.4 Discussion

Beyond pointwise precision, the paper's framing requires that GART 2.0 rank instances by tour cost in agreement with the true ranking. Rank correlations on the same predictions used above confirm this: GART 2.0 reaches Spearman $\rho = 0.9993$ / Kendall $\tau = 0.981$ on the 2D benchmark, $\rho = 1.0000$ / $\tau = 0.998$ on the multi-dimensional set, and $\rho = 0.9991$ / $\tau = 0.985$ on TSPLIB EUC_2D, outranking MST Ratio, GART 1.0, and Hilbert on both metrics in every case. The precision-first design is therefore also rank-preserving, which is what the monotone framing introduced in the Abstract and Section 1 requires of an oracle intended for downstream optimization pipelines.

Global rank correlation near unity is compatible with local pairwise inversions in the dense part of the cost distribution, so we additionally report ε-close pairwise ranking accuracy—the fraction of instance pairs (A, B) with $|L_A - L_B| / \max(L_A, L_B) < \varepsilon$ for which the estimator ranks A, B in the same order as the true optima. At $\varepsilon = 5\%$ GART

2.0 reaches 70.9% on TSPLIB EUC_2D (55 qualifying pairs after the $\alpha \leq 2.5$ filter), 72.8% on the 2D synthetic suite (4.5×10^4 pairs), and 89.9% on the multi-dimensional set (2.3×10^4 pairs), versus 61.8%, 71.0%, and 71.7% respectively for the fixed-MST-ratio baseline. At $\varepsilon = 10\%$ these rise to 78.3%/82.0%/94.7%, again dominating the baseline in every slice. This pairwise-local metric is the signal a downstream planner sees when it must choose between two similar-cost candidates, and it is the one on which a global correlation near unity gives the least information; GART 2.0 leads the baseline on it uniformly, which we consider the operational content of the monotone-oracle framing.

This rank agreement is consistent with a classical asymptotic result that also explains when the fixed MST-ratio baseline catches up. For i.i.d. points of bounded density f in $\mathbb{R}^d$, Beardwood et al. [1959] prove that $n^{-(d-1)/d} L_{\text{TSP}}(n) \to \beta(d) \int f^{(d-1)/d}$ almost surely, and Steele [1988] proves the analogous limit for the MST with its own constant $c(1, d)$ (the Steele theorem states $c(\alpha, d)$ with edge-weight exponent $\alpha = 1$ corresponding to the MST); dividing the two gives $L_{\text{TSP}}(n)/L_{\text{MST}}(n) \to \beta(d)/c(1, d)$ almost surely. In the $d = 100$ bucket MST Ratio already sits at 6.55% MAPE against GART 2.0's 1.04%, and in the largest TSPLIB bucket ($n > 400$) MST Ratio (3.92%) matches GART 2.0 (3.75%) on MAPE with essentially overlapping SDPE bootstrap intervals — exactly the asymptotic regime in which the Beardwood–Steele ratio ceases to vary meaningfully with instance topology. GART 2.0's advantage is concentrated in the mid-density regime where finite-n deviations from the asymptote dominate, which is also where practical logistics instances live.

The $n > 400$ TSPLIB bucket also exhibits $R^2_\alpha = -0.549$, which deserves direct acknowledgement rather than a footnote. The coefficient of determination is computed identically in both forms we audit—`sklearn.metrics.r2_score` and the manual expression $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$—and the two agree to six decimal places, so the sign is not a bug. A negative value is legitimate under the sklearn convention whenever the estimator does worse than the per-bucket mean on α, and three ingredients conspire to push R^2_α below zero on this slice: (i) real-world EUC_2D α concentrates very tightly at large n, $\alpha \in [1.01, 1.20]$ with $\sigma(\alpha) \approx 0.044$, so the denominator SS_{tot} is small; (ii) the model is pulled toward the synthetic-training prior whose mean α at matched n is higher, producing a persistent upward offset; and (iii) the offset is monotone in n—mean $(\hat{\alpha} - \alpha)$ grows from $+0.006$ at $n \leq 100$ to $+0.039$ at $n > 400$, with the fraction of over-predictions rising from 75% to 87% over the same range. The behaviour is therefore a systematic distribution-shift bias between synthetic training and real-world TSPLIB geometry, not a variance problem: the SDPE column, which measures the tightness of the error distribution, remains the tightest in the table in every bucket including $n > 400$, because a consistent offset distorts α estimates far less than it distorts the absolute-ratio target R^2_α. Cost-level accuracy ($R^2 = 1.000$, MAPE 3.75%) is unaffected because L_{MST} dominates L_{TSP} at large n and the multiplicative bias in α becomes a small additive tilt on tour cost. A future revision could correct the offset with a per-n calibration layer; we note the limitation explicitly here rather than hide it in the table.

The classical geometric estimators — BHH, Cavdar, Chien, Daganzo — are absent from the ND results by construction: they require either the convex hull area or the bounding-box volume, and hull computation scales as $O(n^{\lfloor d/2 \rfloor})$ [O'Rourke, 1998] while the hull-to-box volume ratio shrinks exponentially with d [Carlsson and Yu, 2024].

The wall-time breakdown reflects the same MST-centric design. On the TSPLIB log — the only timing log that cleanly separates preparation from inference — GART 2.0 spends 84.8% of its wall time on feature extraction and MST construction and 14.1% on LightGBM inference, while classical baselines spend $\geq 99\%$ on feature and MST preparation. The overhead GART 2.0 introduces relative to any MST-based baseline is therefore the feature pipeline itself, not the tree ensemble; that pipeline cost is sunk as soon as the MST must be built. The remaining weak points are grid lattices and near-linear manifolds (Line Noise) where MST topology becomes structurally degenerate; both configurations are rare in real-world logistics but informative for mapping the model's boundary.

6 Application: Non-Euclidean Estimation via MDS Embedding

The preceding benchmarks evaluate GART 2.0 on Euclidean point clouds, where the MST is built from pairwise distances between coordinates. Many real-world distance matrices are not given as coordinates at all, or are given in a non-Euclidean metric whose native geometry defeats the feature pipeline. This section extends the estimator to those cases through a multidimensional-scaling pre-step that produces a Euclidean embedding consumable by the same trained booster, and reports results on the non-EUC_2D subset of TSPLIB95 where no classical 2D estimator is even defined.

TSPLIB95 contains 32 metric instances that are not native EUC_2D: 4 CEIL_2D (Euclidean with ceiling-rounded distances), 2 ATT (pseudo-Euclidean), 10 GEO (great-circle geographic), and 16 EXPLICIT-metric (distance-matrix only). Classical estimators cannot process these: BHH and Cavdar require 2D coordinates, Chien requires convex hull area, and Hilbert requires a Euclidean embedding. We present GART 2.0 as the sole estimator viable on these cases. Timings are reported as an exploration of capability, not a comparison. We exclude the EXPLICIT outlier `brg180` from every aggregate because its distance matrix has 85,368 triangle-inequality violations: GART 2.0 (like any MST-based estimator) assumes symmetric metric distances, so non-metric instances are out of scope for this model.

6.1 MDS preprocessing

The following procedure converts any TSPLIB distance matrix into a Euclidean coordinate set that the GART 2.0 feature pipeline can consume unchanged. For each non-Euclidean TSPLIB instance we obtain a Euclidean coordinate representation by applying classical (metric) multidimensional scaling [Torgerson, 1952, Gower, 1966, Borg and Groenen, 2005, Cox and Cox, 2000] to the squared distance matrix $D^{(2)}$, double-centring it and taking the leading eigenvectors of the resulting Gram matrix [Gower, 1966]. We use the SciPy eigendecomposition routine [Virtanen et al., 2020] and retain the smallest number of dimensions k that preserves $\geq 99.9\%$ of the positive eigenvalue mass (capped at $k = 100$). Geometric features are computed on the MDS embedding while MST features are computed from the *original* D so no metric information is lost, and the selected k is passed to the model as the `dimension` feature. Full derivation and the per-instance k distribution are given in Appendix C.

6.2 Results on non-Euclidean TSPLIB95

Table 11: GART 2.0 on non-Euclidean TSPLIB95 (32 instances).

Distance type	N	MAPE (%)	SDPE (%)	Median (%)	r_α	Time (ms)	Embed. k (avg [range])
CEIL_2D	4	6.49	6.12	6.15	+0.656	211	2 [2,2]
ATT	2	4.65	0.48	4.65	—	23.4	53 [6,100]
GEO	10	3.92	2.18	4.57	+0.982	9.3	16 [2,61]
EXPLICIT-metric	16	6.22	5.90	5.46	+0.868	7.6	37 [8,100]
TOTAL	**32**	**5.43**	**5.56**	**4.99**	**+0.868**	**9.2**	—

The embedded dimension k is chosen adaptively per instance by the MDS variance-retention rule, and r_α is undefined for the ATT row because only two instances are available. No CONCORDE or LKH-3 solver time is reported: neither solver is natively applicable to non-Euclidean distance types without format-specific translation, which falls outside the scope of this experiment. GART 2.0 generalizes across all four distance types at an aggregate SDPE of 5.56% and MAPE of 5.43% over the 32 instances, with the per-type MAPE spanning 3.92–6.49%. GEO instances achieve the lowest error (MAPE 3.92%) because great-circle distances admit a low-stress Euclidean embedding at typical geographic scales. EXPLICIT-metric instances reach 6.22% because their matrices may mildly violate the triangle inequality, inflating MDS stress. The 211 ms CEIL_2D row is the median over $N = 4$ instances, two of which exceed $n = 10{,}000$; with such a small sample the median is a point estimate, not a distribution, and the two large instances pull it upward.

7 Conclusion

We presented GART 2.0, a TSP tour-length estimator that predicts the MST-to-TSP scaling factor α using 30 dimension-agnostic features combining MST topology with lightweight coordinate-range descriptors. Three benchmarks—2,580 synthetic 2D instances, 16,920 multi-dimensional test instances spanning $d = 2$ to $d = 100$, and 110 metric TSPLIB95 instances (78 EUC_2D plus 32 non-Euclidean via MDS)—demonstrate that MST-topology features are sufficient to recover the per-instance scaling factor with high *precision* and *monotonicity* across dimensions and distribution types, without any enclosure geometry. The primary contribution is not pointwise optimality but a **monotone, precision-first** estimator: one whose SDPE is materially tighter than the MST-ratio baseline and whose error ordering tracks the true cost ordering, so that it can be used as a rank-preserving oracle inside a larger optimization pipeline where a precise-but-biased predictor can be recalibrated through a single scaling constant and a rank-correct predictor drives instance-selection and partitioning decisions.

7.1 Contributions

Across the 2D diverse benchmark, the multi-dimensional synthetic set (d up to 100), and the 78 EUC_2D TSPLIB95 instances (with n reaching 18,512, more than $18\times$ the training cap), GART 2.0 is materially tighter in precision than the MST-ratio baseline, as summarised by the SDPE columns and their 95% bootstrap confidence intervals in the benchmarking tables. On the 32 non-Euclidean TSPLIB95 instances (CEIL_2D, ATT, GEO, EXPLICIT-metric), to our knowledge GART 2.0 is the first tour-length estimator evaluated across every TSPLIB95 metric distance type through a single MDS-hybrid pipeline. The Delaunay-accelerated feature pipeline yields sub-500 ms prediction at $n = 1{,}000$.

7.2 Limitations and Future Work

GART 2.0 as delivered still has three specific limitations, each pointing at a concrete next step: the α regressor is not yet tight, feature extraction time scales with d, and there is an asymptotic regime where a fixed MST-ratio constant already suffices.

1. **The α regressor is not yet tight.** The coefficient of determination on α and related diagnostics (reported in the benchmarking tables) show that a non-trivial share of the variance in the target is still unexplained. The current feature set is a pragmatic first cut; a more expressive regressor (deeper LightGBM, a small neural head on top of the current features, or a probabilistic model targeting the conditional distribution of α rather than its mean) should reduce SDPE further without requiring new feature engineering.
2. **Feature extraction cost scales with d.** The feature set itself is dimension-agnostic (defined and applied without change across all tested d), but feature computation time is not: pairwise distances, centroid statistics, and MST construction all scale with d. The TSPLIB and ND tables show per-instance wall time growing with d even when statistical accuracy does not degrade. A principled dimension-reduction front-end (learned embedding, random projection, or adaptive feature sampling) would make dimension-independence hold in *cost* as well as in *definition*.
3. **One regime already admits the MST-ratio baseline.** At large n under near-uniform spatial density, the true α concentrates tightly around its asymptotic BHH value. In this regime a constant-α estimator is comparable in accuracy to GART 2.0 while costing less to evaluate. GART 2.0's precision advantage is strongest for lower n and for distributions that deviate from uniform (clustered, anisotropic, non-Euclidean via MDS); a natural follow-on is a dispatch layer that runs the cheap MST-ratio baseline when the instance is detected to be in the easy regime and defers to GART 2.0 otherwise.

The natural next step for *use* of the estimator is to embed it as a cost oracle inside a larger optimization loop: a Vehicle Routing Problem sub-tour estimator [Mariescu-Istodor and Fränti, 2021], or a bounding function in branch-and-bound. Both settings would measure whether GART 2.0's precision translates into faster, more reliable end-to-end decisions.

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A Code and Data Availability

Our implementation, trained model weights, and benchmark scripts are available at `https://github.com/sfeldmanMIG25/Area-and-Distribution-Free-Estimator-for-TSP`. Synthetic instances can be regenerated using the provided scripts with fixed random seeds.

B Training Data Generation Details

The training dataset was constructed by generating synthetic TSP instances across four distribution classes, with dimensions independently assigned per axis:

- **Uniform**: Coordinates drawn from $\mathcal{U}(0, G)$ where G is the grid size.
- **Normal (Gaussian)**: Coordinates drawn from $\mathcal{N}(G/2, G/6)$, truncated to $[0, G]$.
- **Clustered (k-means)**: $k \in [3, 15]$ cluster centers placed uniformly, with points drawn from $\mathcal{N}(\mu_k, \sigma_k)$ around each center.
- **Correlated**: Autoregressive structure across dimensions: $x_{d+1} = \rho \cdot x_d + \epsilon$, producing axis-aligned anisotropy.

Each instance's dimensions are independently assigned a distribution class, producing anisotropic configurations where topology varies by axis. Node counts span $n \in [5, 1{,}000]$ with training-visible dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ (17 dimension values), plus a held-out extrapolation bucket at $d = 100$ (4,574 instances, all placed in the test split). Ground-truth solutions were computed by running both Concorde and LKH-3, retaining the shorter result.

The 106,272 solved instances are partitioned into training (69,768, 65.6%), validation (19,584, 18.4%), and test (16,920, 15.9%) using a stratified sample on $(d, n, \text{grid size})$. The stratification rule ranks instances within each stratum by a seeded uniform and assigns roughly the first 16% to test, the next 18% to validation, and the remainder to training; $d = 100$ instances bypass this rule and are locked to the test partition (see `feature_creator_v3.create_stratified_split`). Table 13 lists the per-(d, n) training-instance count; grid size $G \in \{100, 1{,}000, 10{,}000\}$ is sampled uniformly within each cell.

B.1 Multi-dimensional benchmark grid

For reference, the full (d, n) count grid of the 16,920-instance ND benchmark:

Table 12: Multi-dimensional synthetic benchmark grid: instance count per (d, n) bucket (16,920 instances total).

$d \setminus n$	$[5, 10]$	$[11, 50]$	$[51, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$	Total
$d = 2$	162	108	135	27	81	135	**648**
$d =$ 3–5	486	324	405	81	243	405	**1,944**
$d =$ 6–10	810	540	675	135	405	675	**3,240**
$d =$ 15–25	486	324	405	81	243	405	**1,944**
$d =$ 30–50	810	540	675	135	405	675	**3,240**
$d = 100$	1,476	984	1,230	246	738	1,230	**5,904**
Total	**4,230**	**2,820**	**3,525**	**705**	**2,115**	**3,525**	**16,920**

Table 13: Training-set instance count per (d, n) cell (69,768 total; grid size $G \in \{100, 1{,}000, 10{,}000\}$ sampled uniformly within each cell; four distribution classes per axis listed in §B).

$d \setminus n$	$[5, 10]$	$[11, 50]$	$[51, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$	Total
$d = 2$	1,026	684	855	171	513	855	**4,104**
$d =$ 3–5	3,078	2,052	2,565	513	1,539	2,565	**12,312**
$d =$ 6–10	5,130	3,420	4,275	855	2,565	4,275	**20,520**
$d =$ 15–25	3,078	2,052	2,565	513	1,539	2,565	**12,312**
$d =$ 30–50	5,130	3,420	4,275	855	2,565	4,275	**20,520**
Total	**17,442**	**11,628**	**14,535**	**2,907**	**8,721**	**14,535**	**69,768**

C MDS Embedding for Non-Euclidean Instances

For TSPLIB instances with non-Euclidean distance types (GEO: geographic coordinates with great-circle distances; EXPLICIT: distance matrix only), we apply classical Multidimensional Scaling (MDS) to obtain a Euclidean coordinate representation. The embedding dimension k is chosen adaptively to retain at least 99.9% of the positive eigenvalue mass, so that the downstream feature pipeline sees the lowest-dimensional representation that preserves the metric.

Given an $n \times n$ distance matrix D, classical MDS computes the eigendecomposition of the doubly-centered matrix $B = -\frac{1}{2} J D^{(2)} J$ where $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ and $D^{(2)}$ is the element-wise squared distance matrix. The embedding coordinates are $X = V_k \Lambda_k^{1/2}$ where V_k and Λ_k are the top k eigenvectors and eigenvalues.

We embed into k dimensions, selecting the smallest k that retains $\geq$99.9% of eigenvalue variance (capped at $k = 100$). For GEO instances, the selected dimension ranges from 2 to 61; for EXPLICIT instances, from 8 to 100. The MST features (edge weights, degrees, diameter) are computed from the *original* distance matrix to preserve metric accuracy, while the geometric features (centroid statistics, bounding volume, aspect ratio) use the MDS embedding. The dimension k is passed to the model as the `dimension` feature.

For GEO instances, the distance function is well-approximated by Euclidean distances at moderate scales, yielding low MDS stress and 3.92% MAPE. For EXPLICIT instances, the distance matrices may violate metric properties (triangle inequality), producing higher stress. After dropping instances that fail the metric-consistency filter from Section 5.2—only `brg180`, whose published optimum exceeds 2 L_{MST} by two orders of magnitude ($L_{\text{TSP}}/L_{\text{MST}} \approx 160$), ruling out any metric interpretation—the remaining 16 EXPLICIT instances achieve 6.22% MAPE.